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NEUROANATOMY: WEEK 1

CENTRAL NERVOUS SYSTEM (CNS)

- **Main Divisions & Core Concepts**
  - Composed of the **Brain and the Spinal Cord.**
  - Serves as the **primary center for the correlation and integration of nervous information**.
  - **Protection Layers:**
    - **Meninges -** A protective system of membranes.
    - **Cerebrospinal Fluid (CSF) -** The fluid where the CNS is suspended.
    - **Bony Armor -** Protected by the skull and the vertebral column.
  - **Cellular Composition:**
    - **Neurons -** Excitable nerve cells and their processes.
    - **Neuroglia -** Specialized support tissues that hold neurons in place.
  - **Interior:**
    - **Gray Matter -** Composed of nerve cells embedded in neuroglia.
    - **White Matter -** Composed of nerve fibers; the white color is derived from the lipid material in myelin sheaths.

PERIPHERAL NERVOUS SYSTEM (PNS)

- Conduct information **to and from the CNS.**
- Relatively unprotected and are **commonly damaged by trauma.**
  - **Cranial Nerves - 12 pairs**
  - **Spinal Nerves - 31 pairs**

AUTONOMIC NERVOUS SYSTEM

- Concerned with the **innervation of involuntary structures** such as the heart, smooth muscle, and glands within the body.
- Distributed to **CNS** and **PNS**.
  - **SYMPATHETIC NERVOUS SYSTEM** - prepares the body for an emergency.
  - **PARASYMPATHETIC NERVOUS SYSTEM** - conserve and restore energy.

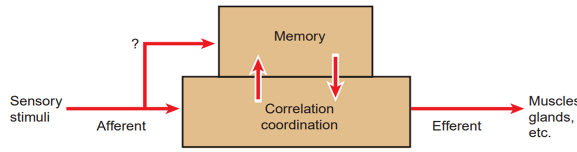


Figure 1-1 The relationship of afferent sensory stimuli to memory bank, correlation and coordinating centers, and common efferent pathway.

MAJOR DIVISIONS OF THE CNS

THE SPINAL CORD

- **Location -** Situated within the **vertebral canal** and is surrounded by 3 meninges:
  - Dura Mater
  - Arachnoid Mater
  - Pia Mater
- Protected by the **Cerebrospinal Fluid** which surrounds the **Subarachnoid space.**
- **Structure - Cylindrical** in shape.
  - Begins at the **Foramen Magnum** (continuous with the medulla oblongata).
  - Terminates at the level of **L1 or L2** in adults (lumbar region)
  - **Conus Medullaris -** The tapered, lower end of the spinal cord.
  - **Filum Terminale -** propagation of Pia Mater from the apex that descends and attaches to the coccyx.
- Attached by 31 pairs of spinal nerves by the **anterior (motor roots) and the posterior (sensory roots).**
- **Posterior Root Ganglion** - cells that give rise to peripheral and central nerve fibers.

- **Composition:**
  - **GRAY MATER** - H-shaped pillar with anterior and posterior gray columns/horns united by a thin gray commissure containing the small central canal.

- **WHITE MATER** - divided into anterior, posterior, and lateral white columns.

BRAIN

- Lies in the **cranial cavity** and is **contagious with the spinal cord through the foramen magnum.**
- Surrounded by **3 meninges; Dura Mater, Arachnoid Mater, Pia Mater.**
- **Cerebrospinal Fluid** - surrounds the brain in the Subarachnoid Space.
- **Major Divisions (Ascending)**
  - **Hindbrain:**
    - **Medulla Oblongata**
      - **Conical**
      - Connects the **pons to the spinal cord**
      - Contains **vital functional nuclei.**
      - Conduit for **ascending and descending nerve fibers.**
    - **Pons -** Located on the anterior surface of the cerebellum, inferior to the midbrain, and superior to the Medulla Oblongata.
      - Contains many nuclei and ascending and descending nerve fibers.
    - **Cerebellum -** Located in the **posterior cranial fossa**; consists of two hemispheres joined by a median portion called the **Vermis**.
      - Connected to the brainstem by **superior, middle, and inferior cerebellar peduncles.**
      - **Cortex** - surface layer of each cerebellar hemisphere.
        - Composed of gray matter.
        - **Dentate Nucleus** - masses of gray matter found in the interior of the cerebellum embedded in the white matter.
      - Voluntary movement planning, initiation, and modification.
      - **Fourth ventricle** - a cavity with cerebrospinal fluid surrounding the cerebellum.
      - Continuous inferiorly to the spinal cord.
  - **Midbrain -** The narrow part connecting the forebrain to the hindbrain.
    - **Cerebral Aqueduct -** narrow cavity of the midbrain.
      - Connects 3rd and 4th ventricles.
- **Forebrain:**
  - **Diencephalon -** almost completely hidden from the surface of the brain.
    - **Thalamus** - sensory relay station.
      - **Large, egg-shaped mass of gray matter** that lies on either side of 3rd ventricle.
      - **Anterior end of Thalamus -** forms the posterior boundary of the **interventricular foramen.**
    - **Hypothalamus** - forms the lower part of the lateral wall and the floor of the 3rd ventricle.
  - **Cerebrum -** The **largest part of the brain.**
    - Consists of two hemispheres connected by the **Corpus Callosum**.
    - The hemispheres are separated by a deep cleft, longitudinal fissure, which

projects the **Falx Cerebri.**

- **Gray Matter:**
  - **Gyri -** Cerebral Folds.
  - **Sulci -** Fissures.
- **White Matter** - containing several huge masses of gray matter, the Basal Nuclei or Ganglia.
  - **Corona Radiata** - fan-shaped collection of fibers.
    - Passes in the white matter to and from the cerebral cortex to the brainstem.
    - Converges on the Basal nuclei and passes between them as the Internal Capsule.
      - **Caudate Nucleus** - tailed nucleus on the ***MEDIAL side*** of the internal capsule.
      - **Lentiform Nucleus** - lens-shaped nucleus on the ***LATERAL side*** of the internal capsule.
  - **Lateral Ventricle** - cavity present within each cerebral hemisphere.
  - Communicates with the 3rd ventricle through **interventricular foramina.**

- **Brainstem -** A collective term for the Medulla Oblongata, Pons, and Midbrain.

CRANIAL NERVES

- **12 pairs**
- Leave the brain and pass through the foramina of the skull.

SPINAL NERVES

- **31 pairs**
- Leave the spinal cord and pass through intervertebral foramina in the vertebral column.
- **Origin -** 8 Cervical, 12 Thoracic, 5 Lumbar, 5 Sacral, 1 Coccygeal.
  - **Roots:**
    - **Anterior Root/Efferent Roots -** Efferent (Motor) fibers carrying impulses away from the CNS.
      - **Motor fibers** - goes to skeletal muscles and cause them to contract.
    - **Posterior Root/Afferent Fibers/Sensory Fibers -** Afferent (Sensory) fibers carrying impulses toward the CNS.
      - Conveys information through **touch, pain, temperature, and vibration.**
    - **Posterior Root Ganglion -** A swelling containing the cell bodies of sensory neurons.

  - **Mixed Nerve -** Formed when the anterior and posterior roots unite at the intervertebral foramen.

- *As the vertebral column grows faster and longer than the spinal cord during development, the spinal nerve roots must lengthen progressively to reach their corresponding exists.*
  - **Upper Cervical** - short roots that run almost HORIZONTALLY.
  - **Lumbar and Sacral** - long roots from a VERTICAL leash of nerves.
    - Together, they are called **Cauda Equina *(latin for horse's tail)***, also known as the **Filum Terminale.**

- **Spinal Nerve Branches (Rami)**
  - **Anterior Ramus -** Supplies muscles and skin of the anterolateral body wall and the limbs.
    - **Plexuses -** Networks formed by anterior rami at the roots of limbs **to form**

complicated nerve plexuses.

        - **Cervical & Brachial -** Upper limb supply.
        - **Lumbar & Sacral -** Lower limb supply.
    - **Posterior Ramus -** Supplies muscles and skin of the back.

- **Ganglia**
    - **Sensory Ganglia/Posterior Ganglia -** Fusiform swellings on the posterior roots of spinal nerves (Spinal Ganglia) or cranial nerves.
    - **Autonomic Ganglia -** Irregularly shaped; situated along the course of efferent fibers of the ANS (e.g., paravertebral sympathetic chains).

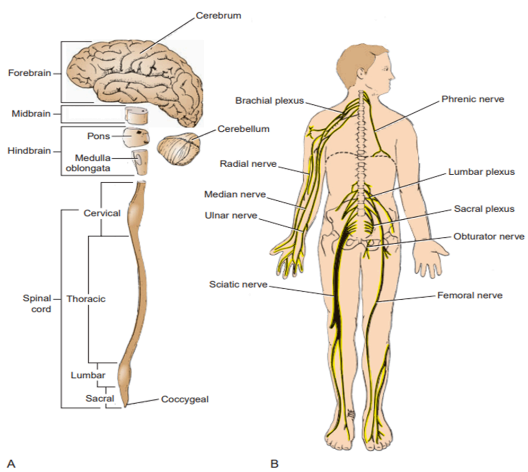
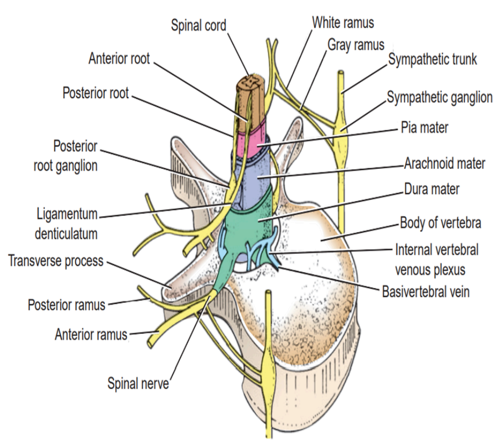
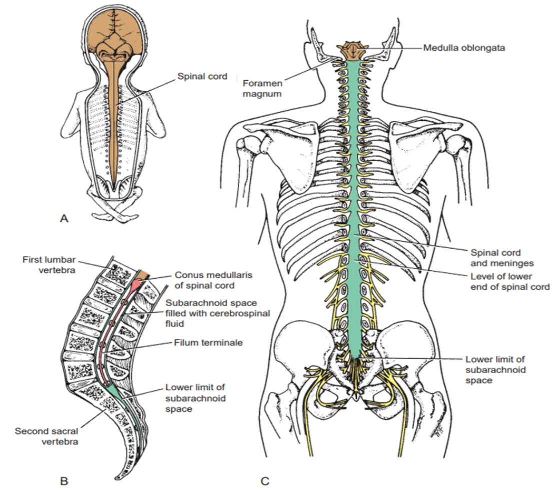
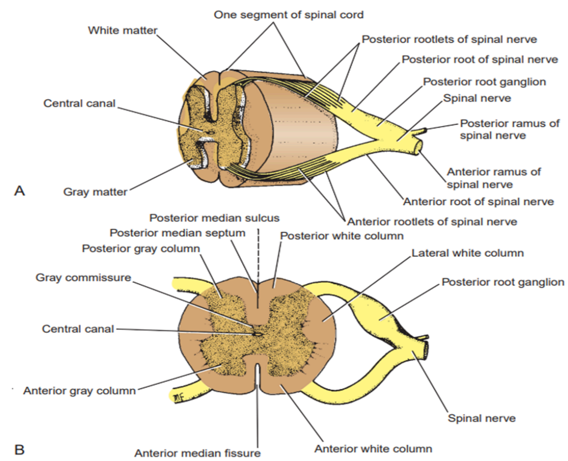
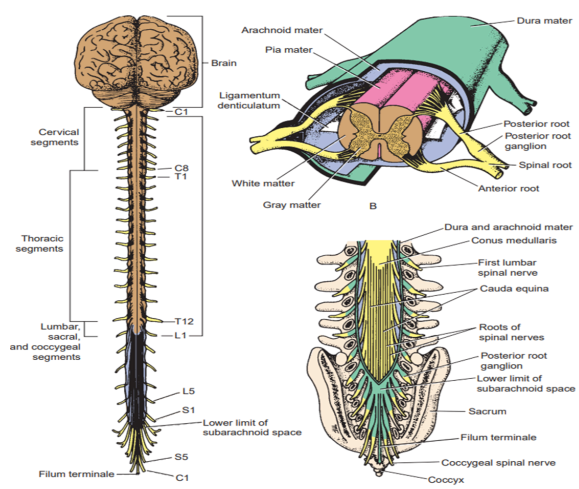


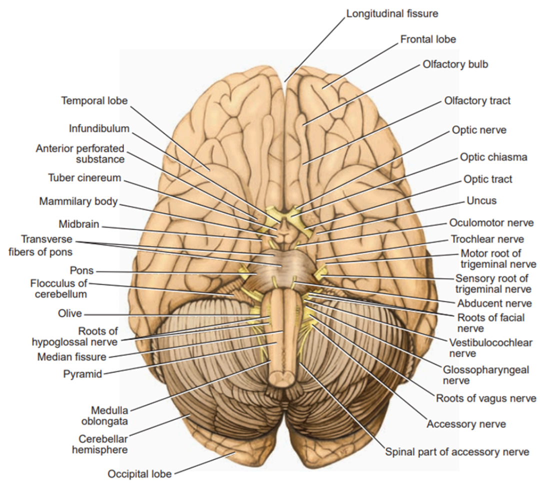
Figure 1-2 **A:** The main divisions of the central nervous system. **B:** The parts of the peripheral nervous system (the cranial nerves have been omitted).

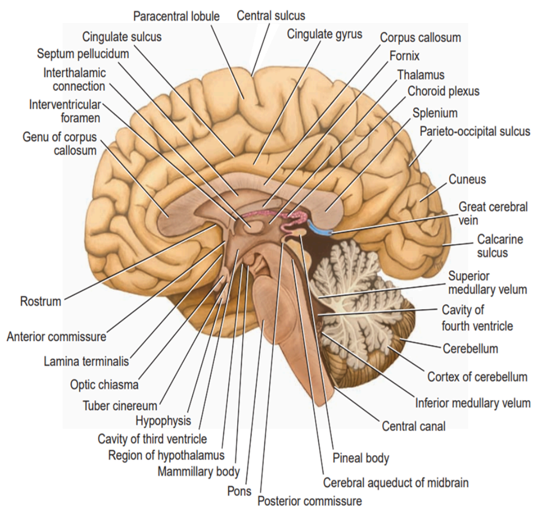




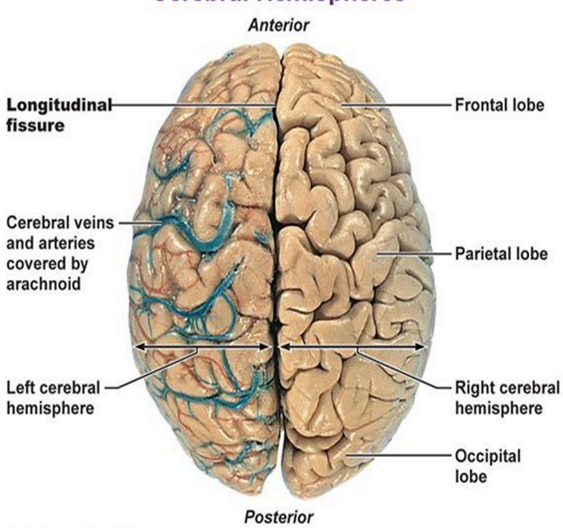




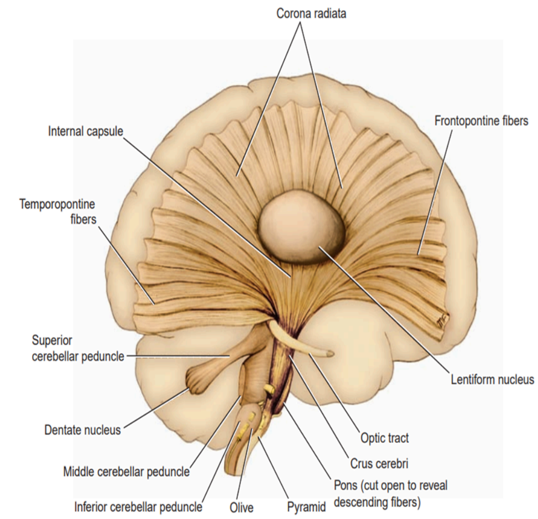


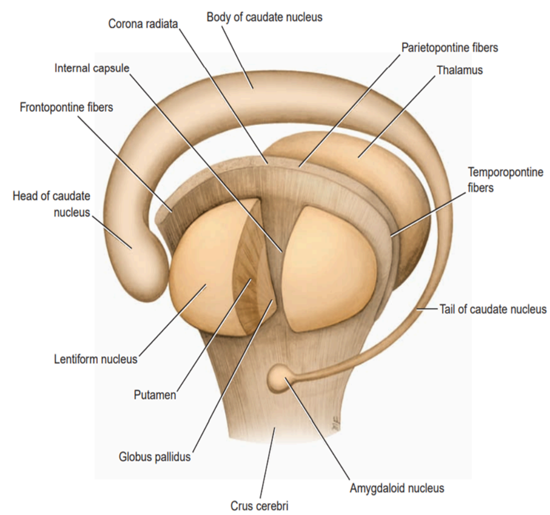


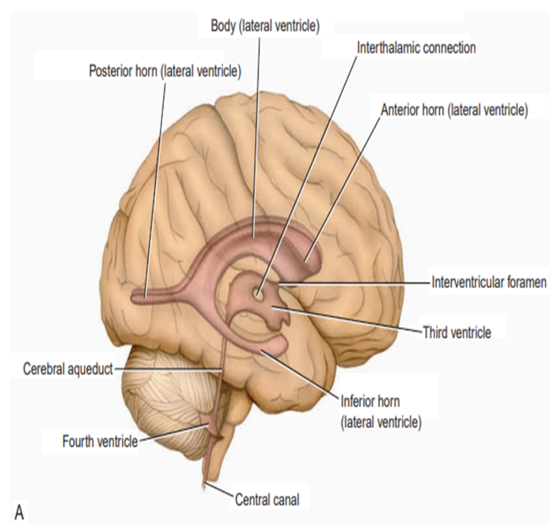
Cerebral Hemispheres

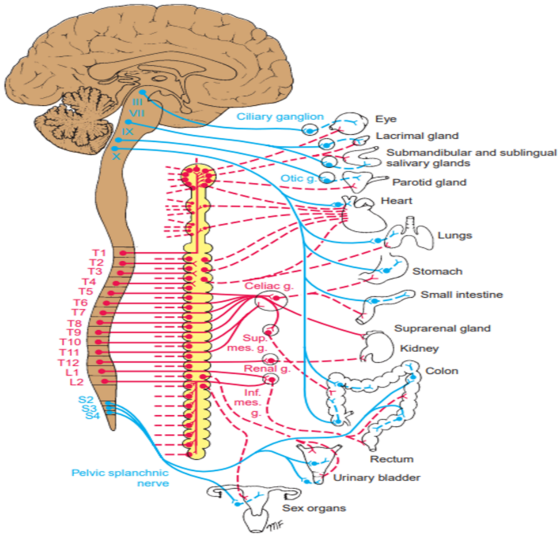


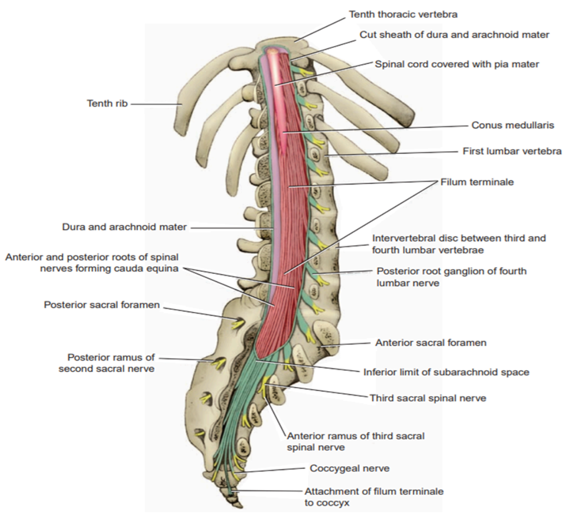
(a) Superior view








III
VII
IX
X
Ciliary ganglion
Eye
Lacrimal gland
Submandibular and sublingual salivary glands
Otic g.
Parotid gland
Heart
Lungs
Stomach
Small intestine
Celiac g.
Suprarenal gland
Sup. mes. g.
Kidney
Renal g.
Colon
Inf. mes. g.
T1
T2
T3
T4
T5
T6
T7
T8
T9
T10
T11
T12
L1
L2
S2
S3
S4
Rectum
Pelvic splanchnic nerve
Urinary bladder
Sex organs


Tenth thoracic vertebra
Cut sheath of dura and arachnoid mater
Spinal cord covered with pia mater
Tenth rib
Conus medullaris
First lumbar vertebra
Filum terminale
Dura and arachnoid mater
Intervertebral disc between third and fourth lumbar vertebrae
Anterior and posterior roots of spinal nerves forming cauda equina
Posterior root ganglion of fourth lumbar nerve
Posterior sacral foramen
Anterior sacral foramen
Posterior ramus of second sacral nerve
Inferior limit of subarachnoid space
Third sacral spinal nerve
Anterior ramus of third sacral spinal nerve
Coccygeal nerve
Attachment of filum terminale to coccyx